

The Tent: Towards a Future of Spatial Entertainment

Movies you can walk around inside of, with or without a headset.

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Figure 1: A volumetric video of an actor discovering a tent in her front yard, the environment is made with photogrammetry. [Photo Credit: MERCANTILE, Rory Mitchell]

ABSTRACT

The dominant form of popular entertainment for the last 125 years, the “rectangle” of film and television, is waning and the emerging medium of Spatial Entertainment is in its infancy. A misplaced focus on the form factor of Virtual Reality headsets has obfuscated this fundamental technological advancement, while creating a bottleneck that has prevented the general population from experiencing this important new medium.

The Tent is an Augmented Reality Narrative, 22 minutes long, built for iPadPro. It allows the audience to explore a 3D world on their tabletop, experiencing the story from any perspective and getting as close to the actor as they wish. Photogrammetry creates photo-real environments and props and a live actor captured with Volumetric Video can convey a more nuanced humanity than a mo-cap or animated actor. (1) Together they are a step towards “movies you can walk around inside of.”

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CCS CONCEPTS

• **Applied computing** → Arts and humanities; Media arts; • **Computing Methodologies** → Computer Graphics; Graphics Systems and interfaces; Mixed / augmented reality; Computer Graphics; Graphics systems and interfaces; Virtual Reality; • **Human-centered computing** → Ubiquitous and mobile computing; Ubiquitous and mobile devices; mobile devices.

KEYWORDS

Spatial Entertainment, Cinema, Television, Theater, Game Engine, Real-Time 3d, Volumetric Video, Photogrammetry

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1 INTRODUCTION

The history of human civilization is the history of humans sharing stories. From the cave paintings in Lascaux and Chauvet to the Ancient Greek theater, from Shakespeare to the Japanese Noh Theater,

from the kinetoscopes, zoetropes, stereoscopes and nickelodeons of the 19th and early 20th centuries to silent films, then talkies, and eventually modern cinema and broadcast television, humans have always adopted the latest technologies to tell their stories. The idea that this evolution should end with the rectangular screen of cinema and television is likely a fallacy. The focus within the entertainment industry on streaming as the primary technological disruptor of the industry misses the emergence of this new medium of Spatial Entertainment.

This isn't a new idea. The French Science-fiction author René Barjavel suggested as much in his book "Total Cinema" in 1944 :

"Since it first emerged, the cinema has been undergoing a constant evolution. This evolution will be complete when it is able to offer us characters in full relief, in full colour, and even perhaps whose perfume we can detect; a time when these characters will be freed from the screen and the darkness of the film theatres to step out into the city streets and the private quarters of their audiences. . . to all intents and purposes it will have reached its ultimate state. A state we call total cinema. " [Leotta, 2018]

Spatial Entertainment will draw upon the tools and techniques of 3000 years of theater, 125 years of cinema, and the last 30 years of real-time, game engine driven 3D Worlds to create a more fluid and exciting form of storytelling. The combination of Volumetric Video and Photogrammetry will allow us to bring the realism and emotional depth of live-action scripted entertainment into the spatial future. We are taking significant strides to resonate with the familiar artistry of the broader entertainment industry while pushing the boundaries of what's technologically and narratively possible.

2 DESIGN AND IMPLEMENTATION

One of the fundamental challenges with Spatial Entertainment is to maintain the visual variety and pacing that modern audiences expect. Our cinematic language of fast-paced editing and montage and lensing breaks the illusion of the physical "presence" of people and objects in a spatial experience. As such, we've adapted a variety of techniques from theater, cinema, and real-time game engines.

Scale and proximity are two of our fundamental spatial tools. They allow us to suggest wide shots and close-ups, a birds-eye-view to emphasize the landscape, or an intimate moment between lovers. Scale and proximity are suggestions; the audience will decide if it's a close-up or a wide-shot, to focus on this actor or that one. It can be exhilarating to be freed from the directorial P.O.V., as though you're an invisible witness to the intimacies of the lives of others.

We've used particle effects and custom shaders from game engines to facilitate scene changes and direct the audiences' attention and we've used theatrical scenography such as a revolving set to move between scenes, re-enforcing the illusion of an object's permanence in the environment by containing its behavior within the laws of physics.

Finally, Volumetric Video is an Actor's medium and performance in volumetric capture allows for traditional acting techniques as well as a more fluid director-actor relationship as seen in Figure 2 and Figure 3.



Figure 2: Rory C. Mitchell directing Maddy Wager in the volumetric studio [Photo Credit: MERCANTILE, Rory Mitchell]



Figure 3: The end result in a photogrammetry environment. [Photo Credit: MERCANTILE, Rory Mitchell]

3 CONCLUSION

Spatial Entertainment could eventually replace or enhance most of the content that is currently consumed in a flat format. Gen Z, raised in the 3D worlds of video games, understands the freedoms of Spatial Entertainment intuitively. Not only entertainment, but sports, news and documentary will be experienced spatially, whether delivered through mobile devices, like *The Tent*, or in XR headsets, holographic tables, the "holodeck" or whatever new delivery methods technology brings. We believe that some of the techniques described here will serve as some of the conceptual underpinnings of a new language of Spatial Entertainment regardless of delivery method. Entertainment will no longer be bound or confined to a rectangle on the wall, or in your hand, but will live in the world around us, on our kitchen tables and in our living rooms, surrounding us, immersing us in the stories that humankind has told from the beginning of time.

REFERENCES

Alfio Leotta, Total cinema: René Barjavel and the future forms of film, *Screen*, Volume 59, Issue 3, Autumn 2018, Pages 372–380, <https://doi.org/10.1093/screen/hjy039>